Joydeep Biswas

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Current Appointment

Associate Professor, Computer Science Department, The University of Texas at Austin
Director, Center for Human and Robot Co-Adaptation (*NSF Science and Technology Center, est. 2026*)
Associate Director, Texas Robotics, The University of Texas at Austin

Visiting Professor, NVIDIA

Education

2014	Ph.D. in Robotics	Carnegie Mellon University
2010	M.S. in Robotics	Carnegie Mellon University
2008	B.Tech. in Engineering Physics	Indian Institute of Technology, Bombay

Achievements and Awards

2026	Chief of Naval Research Fellow, Office of Naval Research, inaugural class
2026	AAAI Senior Member, Class of 2026
2026	IEEE Senior Member
2026	Best Paper Award Finalist in Field and Service Robotics, ICRA 2026, *GuideTWSI*
2026	Best Paper Honourable Mention, HRI 2026, *GuideNav*
2025	Third Place, RoboCup 2025 Humanoid League (Adult Size), *UT Austin Villa*
2025	Best Paper Runner-up, RSS 2025 Workshop on Resilient Off-road Autonomous Robotics, *OVerSeeC*
2025	Best Paper Award, ICRA 2025 Workshop on Advances in Social Navigation, *HEIGHT*
2024	Best Paper Award, ACM CHI Conference on Human Factors in Computing Systems, *Informed Guide Dog Robot Design*
2024	JP Morgan Faculty Research Award
2023	Best Paper Award Nominee, AAAI Spring Symposium on HRI in Academia and Industry: Bridging the Gap, *Benchmarking Social Robot Navigation Across Academia and Industry*
2022	1st Place, Benchmark Autonomous Robot Navigation (BARN) Challenge, ICRA 2022
2021	NSF CAREER Award
2019	Student Best Poster Award, Northrop Grumman University Symposium
2019	IJCAI Early Career Spotlight
2018	Amazon Research Award
2018	JP Morgan AI Faculty Research Award
2018	Best Interactive Demo Award, AAMAS 2018, *Interactive Robot Transition Repair With SMT: Demonstration*
2018	5th place, RoboCup 2018 Small Size League, *UMass MinuteBots*, Faculty Team Leader
2017	Lower Bracket 1st place, RoboCup 2017 Small Size League, *UMass MinuteBots*, Faculty Team Leader
2015	Siebel Scholar, Class of 2015
2015	1st place, RoboCup 2015 Small Size League, *CMDragons*, Student Team Leader

2014 2nd place, RoboCup 2014 Small Size League, *CMDragons*, Student Team Leader
2013 2nd place, RoboCup 2013 Small Size League, *CMDragons*, Student Team Leader
2010 2nd place, RoboCup 2010 Small Size League, *CMDragons*, Team Member

Employment History

2024 - Present	Visiting Professor	NVIDIA
2023 - Present	Associate Professor	Computer Science Department, University of Texas at Austin, TX, USA
2019 - 2025	Adjunct Assistant Professor	College of Information and Computer Sciences, University of Massachusetts Amherst, MA, USA
2019 - 2023	Assistant Professor	Computer Science Department, University of Texas at Austin, TX, USA
2017 - 2023	Technical Advisor	Consumer Robotics, Amazon Lab 126
2015 - 2019	Assistant Professor	College of Information and Computer Sciences, University of Massachusetts Amherst, MA, USA
2015	Post-Doctoral Fellow	Computer Science Department, Carnegie Mellon University, Pittsburgh, PA, USA
2012	Summer Intern	Google Research, Mountain View, CA, USA
2010	Summer Intern	Intel Research, Pittsburgh, PA, USA

Funding

Federal Funding

NSF Science and Technology Center "*Center for Human and Robot Co-Adaptation*"
Role: PI and Center Director.
Award: 2535195.
Period: September 2026 – August 2031.
Five-year, $30M NSF Science and Technology Center.
Six-university collaboration: UT Austin (lead), MIT, Yale University, Indiana University Bloomington, University of Utah, and Tufts University.

NIH Award "*Human-Centered Development of Guide-dog Robots* "
Role: Co-PI.
PI: Donghyun Kim (UMass Amherst).
Co-PIs: Ivan Lee (UMass Amherst), Nicholas Giudice (University of Maine).
Period: October 2025 – September 2028.

NSF Award "*Collaborative Research: SLES: No Bad Surprises: Aligning Agent and Human Norms via Specification Refinements*"
Role: Co-PI.
PI: Sandhya Saisubramanian (Oregon State University).
Co-PIs: Houssam Abbas (Oregon State University), Shlomo Zilberstein (UMass Amherst).
Period: October 2024 – September 2028.

Army Research Laboratories Award "*SARA Sprint 3: Long-Range Offroad Introspective Driving*"
Role: PI.
Period: January 2024 – December 2025.

NSF Award "*FMitF: Track I: Program Synthesis for Robot Learning from Demonstrations*"
Role: Co-PI.
PI: Isil Dillig.
Period: October 2023 – September 2027.

NSF Award "*GCR: Community-Embedded Robotics: Understanding Sociotechnical Interactions with Long-term Autonomous Deployments* "
Role: Co-PI.
PI: Luis Sentis.
Co-PIs: Elliott Hauser, Justin Hart, Keri Stephens.
Period: October 2022 – September 2027.

Army Research Laboratories Award "*Human-Guided Learning of Neuro-Symbolic Mission Execution Policies*"
Role: PI.
Co-PI: Isil Dillig.
Period: September 2021 – January 2023.

NSF Award "*NRT-AI: Convergent, Responsible, and Ethical Artificial Intelligence Training Experience for Roboticists*"
Role: Co-PI.
PI: Junfeng Jiao.
Co-PIs: Luis Sentis, Justin Hart.
Period: September 2021 – August 2026.

NSF Award "*CAREER: Robust Perception and Customization for Long-Term Autonomous Mobile Service Robots*"
Role: PI.
Period: April 2021 – March 2026.

NSF Award "*RI: Medium: Introspective Perception and Planning for Long-Term Autonomy*"
Role: PI.
Co-PI: Shlomo Zilberstein (UMass)
Period: July 2020 – June 2023.

NSF Award "*SHF: Small: Interactive Synthesis and Repair For Robot Programs*"
Role: Co-PI.
PI: Arjun Guha (UMass)
Period: June 2020 – May 2023.

DARPA Award "*Advancing Learning via Probabilistic Causal Analysis for Competency Awareness*"
Role: Co-PI.
PI: Charles River Analytics Co-PI: David Jensen (UMass).
Period: October 2019 – September 2022.

Army Futures Command Robotics Center of Excellence "*Persistent Fully Autonomous Multi-Robot Tactics in Complex Environments*"
Role: Co-PI.
PI: Peter Stone Co-PIs: Luis Sentis, Justin Hart
Period: October 2019 – December 2022.

NSF Award "*S&AS: FND: Reliable Semi-Autonomy with Diminishing Reliance on Humans*"
Role: Co-PI.
PI: Shlomo Zilberstein (UMass)
Period: September 2017 – August 2020.

DARPA Award "*Intelligent Model-Based Adaptation for Mobile Robotics*"
Role: Co-PI.
PI: Jonathan Aldrich (CMU) Co-PIs: David Garlan (CMU), Manuela Veloso (CMU), Christian Kaestner (CMU), Claire Le Gouess (CMU).
Period: November 2015 – November 2019.

Competitive Industry Awards

Amazon Faculty Research Award, 2024
Role: PI.
Collaborators: Roberto Martin-Martin
Period: September 2025 – August 2026.

JP Morgan Faculty Research Award, 2023
Role: PI.
Collaborators: Arjun Guha (Northeastern University)
Period: September 2023 – August 2024.

Northrop Grumman Mission Systems' Research in Applications for Learning Machines (REALM) Consortium
Role: Co-PI.
PI: Shaoshuai Mou (Purdue) Co-PIs: Daniel A. DeLaurentis (Purdue), Bing Liu (UIC)
Period: January 2019 – December 2021.

JP Morgan AI Research Award, 2019
Role: PI.
Period: September 2019 – August 2020.

Amazon Research Award, 2018
Role: PI.
Period: September 2019 – August 2020.

Teaching Experience

Instructor, CS 304I, Fall 2025, Fall 2026: The Essentials of AI for Life and Society
University-wide course, University of Texas at Austin

Instructor, CS 393R, Fall 2020, Fall 2021, Spring 2024: Autonomous Robots
Graduate course, University of Texas at Austin

Instructor, CS 109, Fall 2023: The Essentials of AI for Life and Society
University-wide course, University of Texas at Austin

Instructor, CS 388U, Fall 2023: Planning, Search, and Reasoning Under Uncertainty
Online MS course, University of Texas at Austin

Instructor, CS 378H, Fall 2023: F1/10 Autonomous Driving – Honors
Honors Undergraduate course, University of Texas at Austin

Instructor, CS378/ME379M/ME397/ECE394J/ECE379K, Spring 2023: Connected Autonomous Electric Vehicles
Undergraduate course, University of Texas at Austin

Instructor, CS 393R, Spring 2023: Planning, Search, and Reasoning Under Uncertainty
Graduate course, University of Texas at Austin

Instructor, CS 378H, Spring 2022: F1/10 Autonomous Driving – Honors
Honors Undergraduate course, University of Texas at Austin

Instructor, CS 378F, Spring 2020, Spring 2021: F1/10 Autonomous Driving
Undergraduate course, University of Texas at Austin

Instructor, COMPSCI 220, Fall 2017, Fall 2018: Programming Methodology
Undergraduate course, University of Massachusetts Amherst

Instructor, COMPSCI 403, Fall 2016, Spring 2018 : Introduction To Robotics
Undergraduate course, University of Massachusetts Amherst

Instructor, COMPSCI 603, Spring 2016, Spring 2017, Spring 2019 : Robotics
Graduate course, University of Massachusetts Amherst

Instructor, COMPSCI 691BR, Spring 2017 : Building A Robot Soccer Team
Graduate Seminar, University of Massachusetts Amherst

Invited Talks

Robots Without Boundaries: Embracing the Unseen and Tackling Novel Tasks
Next G Alliance Technology Roadmap Working Group, virtual, August 12, 2026

Habits, Goals, and Surprises: Learning and Reasoning for Open-World Navigation
Keynote talk, RSS Workshop on Open World Navigation in the Foundation Model Era: Robustness and Failure Recovery, Sydney, Australia, July 17, 2026

Robots Without Boundaries: Embracing the Unseen and Tackling Novel Tasks
Machine Learning, Responsible AI, and Robotics Symposium, UT Austin, March 3, 2026

Reading the Dirt: Learning What Matters (and What Doesn't) for Off-Road Navigation
Keynote talk, RSS 2025 Workshop on Resilient Off-road Autonomous Robotics, June 2025

Robots Without Boundaries: Embracing the Unseen and Tackling Novel Tasks
Amazon Consumer Robotics Symposium, May 2025

Robots Without Boundaries: Embracing the Unseen and Tackling Novel Tasks
Keynote talk, Texas Regional Robotics Symposium, May 2025

Teaching Robots To "Get It Right"
Invited talk, Oregon State University, January 2025

Teaching Robots To "Get It Right"
Keynote talk, The 37th International FLAIRS Conference, May 2024

From Social- To Everything- Navigation
Workshop on Social Robot Navigation: Advances and Evaluation. IROS 2023, October 2023

Towards Context-Aware Robot Navigation
Workshop on Robotic Perception and Mapping: Frontier Vision & Learning Techniques: Advances and Evaluation. IROS 2023, October 2023

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous
University of Maryland, October 2022

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous
Northeastern University, October 2022

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous
Wellesley College, October 2022

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous
Samsung AI Center, NYC, June 2022

Self-Supervised and User-Supervised Adaptation of Autonomous Robots
JP Morgan AI Research Center, NYC, June 2022

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous
Stanford University / Robotics Seminar, April 2022

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous
University of Southern California / CS Colloquium, April 2022

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous
Brown University / BigAI Talk, April 2022

Motion Control and Visual Representation Learning for High-Speed Off-Road Driving
University of Pennsylvania / F1Tenth Invited Lecture, April 2022

Particle Filters for Mobile Robot Localization
Wellesley College, February 2022

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous
Nvidia, March 2021

Anticipating and Avoiding Failures Using Introspective Perception and Physics-Informed Program Synthesis
MIT Embodied Intelligence Seminar, February 2021

The Quest for "Always-On" Autonomous Mobile Robots
IJCAI 2019 Early Career Spotlight Talk, August 2019

Building Robots For Long-Term Autonomy, And Keeping Them Autonomous
Yale University, April 2019

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous
ICRA 2018 Workshop: Long-term Autonomy and Deployment of Intelligent Robots in the Real-world, May 2018

Building Robots For Long-Term Autonomy, And Keeping Them Autonomous
Carnegie Mellon University,March 2018

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous
Amazon, November 2017

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous
IROS 2017 Workshop: Assistance and Service Robotics in a Human Environment, September 2017

Autonomous Mobile Robot Perception for Changing Environments
ICRA 2016 Workshop: AI for Long-term Autonomy, May 2016

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous.
University of New Hampshire Robotics Seminar Series, March 2016

The Quest for Robust, Reliable, Autonomous Mobile Robots.
Williams College Computer Science Department Colloquium, November 2015

The Quest for Robust, Reliable, Autonomous Mobile Robots.
Vecna Robotics, September 2015

The Quest for Robust, Reliable, Autonomous Mobile Robots.
University of Minnesota, Computer Science & Engineering, April 2015

Vector Map-Based, Non-Markov Localization for Long-Term Deployment of Autonomous Mobile Robots
Google X, April 2015

The Quest for Robust, Reliable, Autonomous Mobile Robots.
University of Massachusetts Amherst, School of Computer Science, March 2015

Panels

Panel Member, 2025 AAAI Presidential Panel on the Future of AI: `https://aaai.org/about-aaai/presidential-panel-on-the-future-of-ai-research/`

Panel Moderator, Ethics Aware Design of AI
2020 Global Analytics Summit: Ethics in AI, Texas McCombs, November 2020

Discussion Panel, Record of Robotics at CMU Part II, A Live Interview with Manuela Veloso.
CMU Record of Robotics Series, October 2020

Last Mile Autonomous Delivery Systems: A Live Webcast Demonstration, and Panel Discussion
UT Good Systems Webinar, September 2020

The Call for an Accelerated Autonomy – Robotics on the Frontlines of a Crisis
Computing In Our New Normal: A UTCS Webinar, May 2020

Panel Chair, Reasoning and Learning in Real-World Systems for Long-Term Autonomy
AAAI 2018 Fall Symposium, October 2018

Professional Service

Outreach Activities

- Science on Screen Series at Amherst Cinema, Amherst MA, 31 October 2018: Presented an introduction to "Christine" within the scientific context of actual self-driving cars. *The Radical Future of Self-Driving Cars*.
- SciTech Cafe, Northampton MA, 23 January 2017: Presented a scientific talk to a general public audience. *"Where am I?" and Other Fundamental Questions Robots Think Long and Hard to Answer*
- HolyokeCodes, Holyoke MA, 8–12 July 2019: Co-Organized with Arjun Guha, a week-long robotics workshop for high-school students with state-of-the-art soccer-playing robots that we used to compete with at RoboCup. We covered the basic robot sense-plan-act control cycle, computational geometry, and simple adversarial planning. Students implemented building blocks of increasingly complex robot behaviors, leading up to a robot soccer tournament.

Significant Service Roles

- Executive Councilor, Association for the Advancement of Artificial Intelligence (AAAI): current term through 2027
- Student Chair, Association for the Advancement of Artificial Intelligence (AAAI): current
- Co-Editor-in-Chief, Elsevier Robotics and Autonomous Systems: July 2026 – present
- Local Chair, Conference on Robot Learning (CoRL): 2026
- Associate Program Chair, AAAI Conference on Artificial Intelligence (AAAI): 2026, 2027
- Senior Member Track Co-Chair, AAAI Conference on Artificial Intelligence (AAAI): 2025
- Associate Editor, The International Journal of Robotics Research: 2023 – present
- Robotics Area Chair, International Joint Conference on Artificial Intelligence (IJCAI): 2023
- Co-Organizer, Texas Regional Robotics Symposium: April 29, 2022
- Associate Editor, Elsevier Robotics and Autonomous Systems: 2019 – June 2026
- RoboCup Federation Trustee: 2021 – Present
- Diversity and Inclusion Co-Chair, AAAI Conference on Artificial Intelligence (AAAI): 2022, 2023
- RoboCup Executive Committee, Small Size League: 2015 – 2021
- Robot Exhibitions Co-Chair, International Joint Conference on Artificial Intelligence (IJCAI): 2021
- RoboCup Symposium Co-Chair: 2020-2021
- Robotics Track Co-Chair, International Conference on Autonomous Agents and Multiagent Systems (AAMAS): 2019
- Associate Editor, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS): 2016

Senior Program Committee

- AAAI Conference on Artificial Intelligence (AAAI): 2024
- AAAI Conference on Artificial Intelligence (AAAI): 2023
- International Joint Conference on Artificial Intelligence (IJCAI): 2022
- International Conference on Autonomous Agents and Multiagent Systems (AAMAS): 2022
- International Conference on Autonomous Agents and Multiagent Systems (AAMAS): 2021
- AAAI Conference on Artificial Intelligence (AAAI): 2020
- Autonomous Robots and Multirobot Systems Workshop (ARMS): 2016

Program Committee / Reviewer

- Autonomous Robots and Multirobot Systems Workshop (ARMS): 2020
- AAAI Symposium on Educational Advances in Artificial Intelligence: 2021
- RoboCup Symposium: 2015, 2016, 2017, 2018, 2019
- AAAI Undergraduate Consortium: 2021
- IEEE/SICE International Symposium on System Integration (SII): 2019
- Robotics: Science and Systems (RSS): 2015, 2016, 2019
- International Symposium on Multi-Robot and Multi-Agent Systems (MRS): 2019
- International Conference on Autonomous Agents and Multiagent Systems (AAMAS): 2017, 2018
- International Conference on Automated Planning and Scheduling (ICAPS) : 2016, 2018, 2021
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS): 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020
- IEEE International Conference on Robotics and Automation (ICRA): 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021
- IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN): 2010
- IEEE Conference on Human-Robot Interaction (HRI): 2016
- International Joint Conference on Artificial Intelligence (IJCAI): 2016
- International Conference on Autonomous Agents and Multiagent Systems (AAMAS): 2016

Journal Reviewing

- IEEE Robotics and Automation Letters (RA-L): 2017, 2018, 2019, 2020, 2021
- IEEE Robotics and Automation Magazine (IEEE-RAM): 2013, 2014, 2015, 2016, 2019
- IEEE Transactions on Robotics (T-RO): 2015, 2018, 2019, 2020
- International Journal of Robotics Research (IJRR): 2016, 2017
- International Journal of Social Robotics (SORO): 2018, 2019

Grant Reviewing

NSF Panelist: 2016(x2), 2018, 2019, 2020, 2021, 2022

University Service

College Level

- Faculty Hiring Committee for Whole Communities Whole Health Cluster Hires: 2019–2020
- CNS Fall Lab Working Group in response to COVID restrictions: 2020

Department Level

- Texas Robotics Machine Shop Committee: 2020–2022
- Texas Robotics Space Committee: 2020–2022
- UTCS Turing Scholars Admissions Committee: 2020–2022
- UTCS Diversity, Equity, and Inclusion Committee: 2020–2022
- UTCS Graduate Admissions Committee: 2019–2020
- UMass CICS Honors Program Director: 2018–2019
- UMass CICS Undergraduate Course Assistant Program Director: 2018–2019
- UMass CICS Graduate Admissions Committee: 2015–2016
- UMass CICS Student Activities Committee: 2015–2016
- UMass CICS Data Science Faculty Hiring Committee: 2016–2017
- UMass CICS Student Activities Committee: 2016–2017

Advising and Thesis Committees

Postdoctoral Supervisor

- Rohan Chandra, 2022 – 2024, currently Assistant Professor at the University of Virginia
- Kiarash (Kia) Rahmani, 2022 – 2024, currently Applied AI Scientist at Durable AI
- Nathan Tsoi, 2025 – present

PhD Supervisor

- Taijing Chen, UT Austin. 2024–present
- Sarah Etter, UT Austin. 2023–present
- Dongmyeong Lee, UT Austin. 2023–present
- Arthur Zhang, UT Austin. 2022–present

- Zichao Hu, UT Austin. 2022–present
- Sadanand Modak, UT Austin. 2022–present
- Eric Hsiung, UT Austin. 2022–present
- Emily Pruc, UMass Amherst. 2018–2022
- Amanda Adkins, UT Austin. 2020–2026, currently at Amazon

- Sadegh Rabiee, UT Austin. 2016–2022, currently at Amazon Lab126
  Introspective Perception for Mobile Robots
- Jarrett Holtz, UT Austin. 2015–2022, currently at Bosch Research
  Leveraging Program Synthesis for Robust Long-Term Robot Autonomy via Interactive Learning and Adaptation
- Samer Nashed (Changed advisors in 2019), UMass Amherst. 2015–2019
- Spencer Lane (Changed advisors in 2019), UMass Amherst. 2016–2019
- Alyxander Burns (Changed advisors in 2019), UMass Amherst. 2017–2019

Master's Thesis Supervisor

- Kavan Sikand, UT Austin. 2019–2022
- David Balaban, UMass Amherst, 2016 – 2018
  A Real-Time Solver For Time-Optimal Control Of Omnidirectional Robots with Bounded Acceleration

Undergraduate Honors Thesis Supervisor

- Elvin Yang, UT Austin. 2021–2022
  Wait, That Feels Familiar: Learning to Extrapolate Human Preferences for Preference-Aligned Path Planning
  Best Honors Thesis Award
- Rahul Menon, UT Austin. 2021–2022
  Terrain-Adaptive Global Planning from Local Demonstrations
- Shakeel Samsudeen, UT Austin. 2021–2022
  Context-Aware Object SLAM
- Nathaniel Plaxton, UT Austin. 2021–2022
  Estimating Kinodynamic Uncertainty Using Learned Gaussian Noise Models
- Michael Satanovski, UT Austin, 2021-2022
  An Empirical Evaluation of LIDAR Object Detectors for Autonomous Mobile Robots
- Edward Schneeweiss, UMass Amherst, 2015 – 2019
  Joint Perception and Planning for Obstacle Avoidance over Non-Planar Terrain
- Kyle Vedder, UMass Amherst, 2015 – 2019
  X: Anytime Multiagent Path Planning With Bounded Search*

- George Larionov, UMass Amherst, 2015 – 2016
  Human-robot Interaction: Integrating Speech Recognition with a Mobile Robot System

Undergraduate Research Awards to Advisees

- Luisa Mao, UT Austin: CRA Outstanding Undergraduate Researcher Award, Awardee (2026); Honorable Mention (2025)
- Jimmy Xin, UT Austin: CRA Outstanding Undergraduate Researcher Award, Honorable Mention (2024)
- Jenna Hammond, Mount Holyoke College (summer research advisee): CRA Outstanding Undergraduate Researcher Award, Honorable Mention (2020)

PhD Committee Member

- Connor Basich, UMass Amherst. Supervisor: Shlomo Zilberstein
- Minkyu Kim, UT Austin. Supervisor: Luis Sentis
- Abhinav Verma, UT Austin. Supervisor: Swarat Chaudhuri
- Kyle Hollins Wray, UMass Amherst. Supervisor: Shlomo Zilberstein
- Justin Svegliato, UMass Amherst. Supervisor: Shlomo Zilberstein
- Tiffany Liu, UMass Amherst. Supervisor: Roderic Grupen
- Takeshi Takahashi, UMass Amherst. Supervisor: Roderic Grupen
- Mike Lanighan, UMass Amherst. Supervisor: Roderic Grupen
- Keen Sung, UMass Amherst. Supervisor: Brian Levine
- (Thesis Opponent) [1], Nils Bore, KTH. Supervisor: John Folkesson

Undergraduate Honors Thesis Committee Member

- Stefan Kussmaul, UMass Amherst. Supervisor: Roderic Grupen
- Karl Schmeckpepper, UMass Amherst. Supervisor: Roderic Grupen

[1] A PhD thesis dissertation in the Swedish doctoral system is formally presented by an external examiner, called the *thesis opponent*. A thesis opponent places the work of the PhD thesis in context with the state of the art, presents the findings of the thesis, and leads a discussion with questions.

Publications

Conference Papers

[1] Aleksander Boruch-Gruszecki, Yangtian Zi, Zixuan Wu, Tejas Oberoi, Carolyn Jane Anderson, Joydeep Biswas, and Arjun Guha. "Agnostics: Learning to Code in Any Programming Language via Reinforcement with a Universal Learning Environment." In: *International Conference on Learning Representations (ICLR)*. 2026. URL: https://arxiv.org/abs/2508.04865.

[2] Taijing Chen, Sateesh Kumar, Junhong Xu, Georgios Pavlakos, Joydeep Biswas, and Roberto Martín-Martín. "Searching in Space and Time: Unified Memory-Action Loops for Open-World Object Retrieval." In: *2026 IEEE International Conference on Robotics and Automation (ICRA)*. 2026. URL: https://joydeepb.com/Publications/icra2026_memory_action.pdf.

[3] Hochul Hwang, Soowan Yang, Jahir Sadik Monon, Nicholas A. Giudice, Sunghoon Ivan Lee, Joydeep Biswas, and Donghyun Kim. "GuideNav: User-Informed Development of a Vision-Only Robotic Navigation Assistant for Blind Travelers." In: *Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction*. 2026, pp. 1129–1139. DOI: 10.1145/3757279.3785625. URL: https://joydeepb.com/Publications/hri2026_guidenav.pdf.

[4] Hochul Hwang, Soowan Yang, Anh H. N. Nguyen, Parth Goel, Krisha Adhikari, Sunghoon Ivan Lee, Joydeep Biswas, Nicholas A. Giudice, and Donghyun Kim. "GuideTWSI: A Diverse Tactile Walking Surface Indicator Dataset from Synthetic and Real-World Images for Blind and Low-Vision Navigation." In: *2026 IEEE International Conference on Robotics and Automation (ICRA)*. 2026. URL: https://joydeepb.com/Publications/icra2026_guidetwsi.pdf.

[5] Dongmyeong Lee, Jesse Quattrociocchi, Christian Ellis, Rwik Rana, Amanda Adkins, Adam Uccello, Garrett Warnell, and Joydeep Biswas. "BEV-Patch-PF: Particle Filtering with BEV-Aerial Feature Matching for Off-Road Geo-Localization." In: *2026 IEEE International Conference on Robotics and Automation (ICRA)*. 2026. URL: https://arxiv.org/pdf/2512.15111.

[6] Wei Liu, Huihua Zhao, Chenran Li, Yuchen Deng, Joydeep Biswas, Soha Pouya, and Yan Chang. "COMPASS: Cross-Embodiment Mobility Policy via Residual RL and Skill Synthesis." In: *2026 IEEE International Conference on Robotics and Automation (ICRA)*. 2026. URL: https://joydeepb.com/Publications/arxiv2025_compass.pdf.

[7] Rwik Rana, Jesse Quattrociocchi, Dongmyeong Lee, Christian Ellis, Amanda Adkins, Adam Uccello, Garrett Warnell, and Joydeep Biswas. "OVerSeeC: Open-Vocabulary CostMap Generation from Satellite Images and Natural Language." In: *2026 IEEE International Conference on Robotics and Automation (ICRA)*. 2026. URL: https://arxiv.org/pdf/2602.18606.

[8] Nathan Tsoi, Eric Hsiung, Masayuki Yamazaki, Swarat Chaudhuri, Peter Stone, and Joydeep Biswas. "Coadaptive Value Alignment." In: *Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS) Blue Sky Ideas Track*. 2026. DOI: 10.65109/naik4475. URL: https://joydeepb.com/Publications/aamas2026_coadaptive.pdf.

[9] Zifan Xu, Kristen Procko, Michael Munje, Kristin Patterson, Lea Sabatini, Joydeep Biswas, and Peter Stone. "The Essentials of AI for Life and Society: A Full-Scale AI Literacy Course Accessible to All." In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 40. 47. 2026, pp. 40730–40736. DOI: 10.1609/aaai.v40i47.41521. URL: https://joydeepb.com/Publications/eaai2026_essentials.pdf.

[10] Zifan Xu, Myoungkyu Seo, Dongmyeong Lee, Hao Fu, Jiaheng Hu, Jiaxun Cui, Yuqian Jiang, Zhihan Wang, Anastasiia Brund, Joydeep Biswas, and Peter Stone. "Learning Agile Striker Skills for Humanoid Soccer Robots from Noisy Sensory Input." In: *2026 IEEE International Conference on Robotics and Automation (ICRA)*. 2026. URL: https://joydeepb.com/Publications/icra2026_striker.pdf.

[11] Shangqun Yu, Hochul Hwang, Trung M. Dang, Joydeep Biswas, Nicholas A. Giudice, Sunghoon Ivan Lee, and Donghyun Kim. "Human-Centered Development of Guide Dog Robots: Quiet and Stable Locomotion Control." In: *2026 IEEE International Conference on Robotics and Automation (ICRA)*. 2026. URL: https://joydeepb.com/Publications/arxiv2025_guide_dog_hcd.pdf.

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